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NURBS Interpolator With Pre-compensation Based on Discrete Inverse Transfer Function for CNC High-precision Machining

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Research Article

Keywords: discrete, transfer function, interpolator with pre-compensation, feedrate scheduling, contour error

Posted Date: August 5th, 2021

DOI: https://doi.org/10.21203/rs.3.rs-768203/v1

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NURBS interpolator with pre-compensation based on discrete inverse transfer function for CNC high-precision machining

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Abstract:

In the field of CNC machining, high-speed and high-precision machining has been regarded as the key research by many scholars. In conventional methods, high-speed machining and high-precision machining are contradictory. It is inevitable to reduce the feedrate to improve the processing accuracy. In the paper, a pre-compensation based on discrete inverse transfer function (PDIT) theory is proposed. PDIT is able to improve machining accuracy by reducing contour errors without decreasing feedrate. The proposed PDIT theory is divided into three parts, NURBS interpolator, feedrate scheduling, and interpolator with pre-compensation. The NURBS interpolator has greatly advantage to interpolate the parameter curve directly. Therefore, the paper adopts the NURBS interpolator to accomplish interpolation. In the feedrate scheduling, S-type flexible acceleration and deceleration are used for path planning, and the maximum starting feedrate is obtained with the feedrate constraint. In the interpolator with pre-compensation, the NURBS interpolator is pre-compensated by PDIT. For inputs, the response of transfer function reach steady-state response with a little time. Before reaching steady-state response, the unsteady response exists in the transfer function. The unsteady response usually sustains tens of interpolation periods and must be lead contour error in machining. Hence, the PDIT theory is employed to compensate the contour error causing by the unsteady response of transfer function to NURBS interpolator. The drive system is a transfer function, so the unsteady response of drive system cause machining errors before reaching the steady-state response. In the paper, the NURBS interpolator is pre-compensated by PDIT theory before the drive system to reduce contour errors and improve machining accuracy. Finally, the performance of the proposed PDIT is evaluated by simulation experiments. The experimental results illustrate that PDIT theory obviously improve the machining accuracy and reduces the contour error.

Keywords: discrete, transfer function, interpolator with pre-compensation, feedrate scheduling, contour error

1. Introduction

In modern manufacturing, the demand for high-speed and high-precision machining is increasing, which has always been a research topic for many scholars. Many scholars reduce the vibration of the machine tool to achieve the purpose of improving the machining accuracy by reducing the fluctuation of the feedrate. Such as, Dong and Wang proposed the Target feedrate filter method to achieve smooth federate [1, 2]. Scholars such as Huan Liu proposed a linear programming mathematical model of the optimal feedrate in the sensitive area in order to obtain the optimal feedrate [3]. But the shortcoming of the method is low real-time performance, which needs to be completed with the help of Matlab linprog. Scholars such as Guo transformed the axial and tangential feedrate constraints into a convex optimization problem in order to find the optimal feedrate [4]. There are also many scholars who improve machining accuracy through optimizing interpolation algorithms. Scholars such as Wolfgang BOHM propose parameter interpolation algorithms based on

Lagrangian [5]. Scholars Sang and Yao propose interpolation algorithms based on Taylor expansion [6, 9]. Scholars Qin Hu propose optimized Taylor Parameter interpolation algorithm to make up for the truncation error [10, 15]. The above scholars are mainly concerned about the CNC system, and some scholars concern about the drive system. Scholars who pay attention to drive system mainly study contour errors caused by drive system. Hu propose an analytical contour compensation algorithm to improve machining accuracy [16, 17]. Scholars such as Li proposed a method based on monocular-vision to obtain and compensate contour errors [18]. Whether the feedrate smoothing algorithm and parameter interpolation algorithm in the CNC system, or the contour error compensation algorithm in the drive system, they are all subjectively divided into two parts. However, there are few research literatures that combine the two together. Therefore, the paper proposes the PDIT theory and makes full use of the relationship between the two to improve machining accuracy without losing feedrate. The PDIT theory studies the influence of transfer function of drive system on contour error, and directly compensates the contour error to the commanded position of CNC.

In order to improve the performance of the NURBS interpolator, PDIT theory is proposed in the paper. PDIT is consist by NURBS interpolator, feedrate scheduling, and interpolator with pre-compensation. NURBS curve has greatly advantage of expressing free curve and is adopted in the paper. In order to obtain the maximum smooth feedrate and improve the efficiency of planning path, the feedrate scheduling based on S-type flexible acceleration and deceleration algorithm are employed. This is introduced in Chapter2. In Chapter3, the principle of PDIT is introduced in detail. PDIT has ability to issue the contour error caused by unsteady response. In the Chapter4, simulation experiments are carried out to evaluate the proposed NURBS interpolator. And conclusion is in the Chapter5.

2. Implementation of feedrate scheduling for NURBS interpolator

Non-uniform rational B-Splines (NURBS) is defined by the node vector U, a set of control points **P** and the weight of each control point. Assuming that the tool path is described by a NURBS curve of order p, the curve can be expressed as [19-21]:

$$C(u)=\frac{\sum_{i=0}^{n} N_{i,p}(u)\omega_i \mathbf{P}_i}{\sum_{i=0}^{n} N_{i,p}(u)\omega_i} \tag{1}$$

where $\mathbf{P}_i$ is the control point, ω_i is the corresponding weights of $\mathbf{P}_i$, $(n+1)$ is the number of control points, and p is the degree of a NURBS curve. $N_{i,p}(u)$ is the p power of B-spline basis functions defined on the non-uniform knot vector $\mathbf{U}=\{u_0,u_1,...,u_{n+p+1}\}=\{\underbrace{a,a,...a}_{p+1},u_{p+1},...u_{m-p-1},\underbrace{b,b,...b}_{p+1}\}$, and $m+1$ is the number of knots. The de Boor-Cox algorithm can be used to calculate the p-order B-spline basic function N_i, $N_{i,k}(u)$ is [19]:

$$\begin{cases} N_{i,0}=\begin{cases}1.....u\in[u_i,u_{i+1}]\\ 0.....otherwise\end{cases} \\ N_{i,k}(u)=\dfrac{u-u_i}{u_{i+k+1}-u}N_{i,k-1}(u)+\dfrac{u_{i+k}-u}{u_{i+k}-u_{i+1}}N_{i+1,k-1}(u) \\ spitulate:\dfrac{0}{0}=0 \end{cases} \tag{2}$$

NURBS interpolator has ability to interpolate parameter curve directly. Compared with the traditional micro-line segment interpolation, NURBS interpolator has outstanding performance in improvement of machining accuracy and efficiency. It is indispensable for high-speed and high-precision machining. Therefore, the paper adopts the NURBS interpolator to accomplish the interpolation, and the relationship between interpolation position u_{i+1} and interpolation position u_i is [20, 23, 25]:

$$u_{i+1} = u_i + \frac{V(u_i)\cdot T}{\left\| \frac{dC(u)}{du} \right\|_{u=u_i}} - \frac{V(u_i)^2 \cdot T^2 \cdot \left(\frac{dC(u)}{du} \cdot \frac{d^2C(u)}{du^2} \right)\Big|_{u=u_i}}{2 \cdot \left\| \frac{dC(u)}{du} \right\|^4_{u=u_i}} \tag{3}$$

It can be observed that to obtain the parameter u_{i+1} of the next interpolation period from formula(3), the feedrate $V(u_i)$ is indispensable. Therefore, the feedrate scheduling is the highlight in the section. The first part is the S-type flexible acceleration and deceleration algorithm, which is the basis of feedrate scheduling. The second part is the feedrate scheduling based on motion model. The third part is the double-direction look-ahead algorithm. The double-direction look-ahead algorithm is divided into backward planning and forward planning. The purpose of backward planning is to obtain the maximum starting speed under the constraint of sensitive speed area. And the purpose of forward planning is using the maximum starting speed to plan the duration time (τ_1, τ_2, τ_3, τ_4, τ_5, τ_6, τ_7) of each segment in the 7-segment S-type acceleration and deceleration for a path with a fixed length of L.

2.1 S-type acceleration and deceleration algorithm

Compared with trapezoidal acceleration and deceleration, the S-type acceleration and deceleration algorithm is more stable and has less impact on the drive system. It is suitable for high-speed and high-precision machining.

A complete S-type acceleration and deceleration has a total of 7 segments [26, 29], named acc-acceleration(AA) segment, uniform acceleration(UA) segment, dec-acceleration(DA) segment, uniform speed(US) segment, dec-deceleration(DD) segment, uniform deceleration(UD) segment and acc- deceleration(AD) segment. Fig. 1 is a complete S-type acceleration and deceleration curve. The details of each segment are described in detail in Appendix A, and the meaning of t_i, T_i, τ_i, $a_i(t)$, a_i, $v_i(t)$, v_i ,$s_i(t)$ s_i, S_{total} is also introduced in Appendix A.

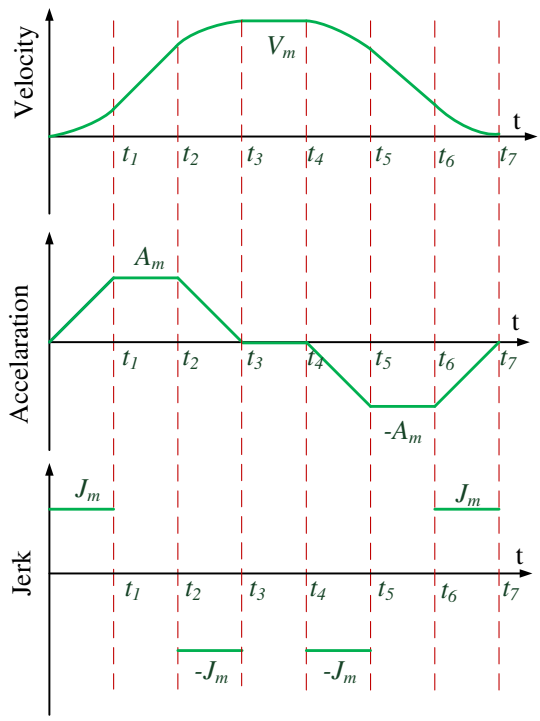

Fig. 1 A complete 7-segment S-type acceleration and deceleration curve. J_m is the maximum jerk, A_m is maximum acceleration and V_m is the maximum velocity

Jerk, acceleration, velocity, and displacement are respectively obtained from formulas (4)(5)(6)(7)for the 7-segment S-type flexible acceleration and deceleration algorithm.

$$J(t)=\begin{cases} J_1(t)=J_m & t_0<t<t_1 & J_1(\mathrm{t}_1)=J_m \\ J_2(t)=0 & t_1<t<t_2 & J_2(\mathrm{t}_2)=0 \\ J_3(t)=-J_m & t_2<t<t_3 & J_3(\mathrm{t}_3)=-J_m \\ J_4(t)=0 & t_3<t<t_4 & J_4(\mathrm{t}_4)=0 \\ J_5(t)=-J_m & t_4<t<t_5 & J_5(\mathrm{t}_5)=-J_m \\ J_6(t)=0 & t_5<t<t_6 & J_6(\mathrm{t}_6)=0 \\ J_7(t)=J_m & t_6<t<t_7 & J_7(\mathrm{t}_7)=J_m \end{cases} \tag{4}$$

$$a(t)=\begin{cases} a_1(t)=J_m\cdot t & t_0<t<t_1 & a_1=a_1(\mathrm{t}_1)=A_m \\ a_2(t)=J_m\cdot T_1=A_\mathrm{m} & t_1<t<t_2 & a_2=a_2(\mathrm{t}_2)=A_m \\ a_3(t)=A_m-J_m\cdot(t-t_2) & t_2<t<t_3 & a_3=a_3(\mathrm{t}_3)=0 \\ a_4(t)=0 & t_3<t<t_4 & a_4=a_4(\mathrm{t}_4)=0 \\ a_5(t)=-J_m\cdot(t-t_4) & t_4<t<t_5 & a_5=a_5(\mathrm{t}_5)=-A_m \\ a_6(t)=-A_m & t_5<t<t_6 & a_6=a_6(\mathrm{t}_6)=-A_m \\ a_7(t)=-A_m+J_m\cdot(t-t_6) & t_6<t<t_7 & a_7=a_7(\mathrm{t}_7)=0 \end{cases} \tag{5}$$

$$\mathrm{v}(t)=\begin{cases} v_1(t)=v_s+\frac{1}{2}J_m(t-t_0)^2 & t_0<t<t_1 & v_1=v_1(t_1)=v_s+\frac{1}{2}J_mT_1^2 \\ v_2(t)=v_1+A_\mathrm{m}(t-t_1) & t_1<t<t_2 & v_2=v_2(t_2)=v_1+A_mT_2 \\ v_3(t)=v_2+A_m(t-t_2)-\frac{1}{2}J_m(t-t_2)^2 & t_2<t<t_3 & v_3=v_3(t_3)=v_2+A_mT_3-\frac{1}{2}J_mT_3^2 \\ v_4(t)=v_3 & t_3<t<t_4 & v_4=v_4(t_4)=v_3 \\ v_5(t)=v_4-\frac{1}{2}J_m(t-t_4)^2 & t_4<t<t_5 & v_5=v_5(t_5)=v_4-\frac{1}{2}J_mT_5^2 \\ v_6(t)=v_5-\mathrm{A_m}(t-t_5) & t_5<t<t_6 & v_6=v_6(t_6)=v_5+a_6T_6=v_5-A_mT_6 \\ v_7(t)=v_6-A_m(t-t_6)+\frac{1}{2}J_m(t-t_6)^2 & t_6<t<t_7 & v_7=v_7(t_7)=v_6-A_mT_7+\frac{1}{2}J_mT_7^2 \end{cases} \tag{6}$$

$$\mathrm{s}(t)=\begin{cases} s_1(t)=v_s(t-t_0)+\frac{1}{6}J_m(t-t_0)^3 & t_0<t<t_1 & d_1=d_1(t_1)=v_0t_1+\frac{1}{2}J_mt_1^3 \\ s_2(t)=s_1+v_1(t-t_1)+\frac{1}{2}A_m(t-t_1)^2 & t_1<t<t_2 & d_2=d_2(t_2)=d_1+v_1T_2+\frac{1}{2}A_mT_2^2 \\ s_3(t)=s_2+v_2(t-t_2)+\frac{1}{2}A_m(t-t_2)^2-\frac{1}{6}J_m(t-t_2)^3 & t_2<t<t_3 & d_3=d_3(t_3)=d_2+v_2T_3+\frac{1}{2}A_mT_3^2-\frac{1}{6}J_mT_3^3 \\ s_4(t)=s_3+v_3(t-t_3) & t_3<t<t_4 & d_4=d_4(t_4)=d_3+v_4T_4 \\ s_5(t)=s_4+v_4(t-t_4)-\frac{1}{6}J_m(t-t_4)^3 & t_4<t<t_5 & d_5=d_5(t_5)=d_4+v_4T_5-\frac{1}{6}J_mT_5^3 \\ s_6(t)=s_5+v_5(t-t_5)-\frac{1}{2}A_m(t-t_5)^2 & t_5<t<t_6 & d_6=d_6(t_6)=d_5+v_5T_6-\frac{1}{2}A_mT_6^2 \\ s_7(t)=s_6+v_6(t-t_6)-\frac{1}{2}A_m(t-t_6)^2+\frac{1}{6}J_m(t-t_6)^3 & t_6<t<t_7 & d_7=d_7(t_7)=d_6+v_6T_7-\frac{1}{2}A_mT_7^2+\frac{1}{6}J_mT_7^3 \end{cases}$$

(7)

To sum up, in the 7 segments of S-type acceleration and deceleration, the analytical formulas of jerk, acceleration, velocity, and displacement have common characteristics. The unified form of the formula is:

$$J(t) = C \tag{8}$$

$$a(t) = J_m \cdot T - J_m \cdot t \tag{9}$$

$$v(t) = v_s + A_m \cdot t + \frac{1}{2} J_m \cdot t^2 \tag{10}$$

$$s(t) = s + v \cdot t - \frac{1}{2} A_m \cdot t^2 + \frac{1}{6} J_m \cdot t^3 \tag{11}$$

2.2 Feedrate scheduling

The process of path planning will not always appear all the 7 segments. Only when the path is long enough, all the 7 segments will exist in the drive system, which is a complete acceleration and deceleration speed curve. If the path is not long enough, not all the 7 segments but parts of them will appear in the drive system. Therefore, it is important to determine which part appears. The detailed analysis is given below.

In the permutation and combination rules of probability theory, the combination of 7 segments are 2^7 different combinations in total. Some permutation and combination results do not exist in practice, even so, there are still many combinations. It is too complex to analyze for each situation. Therefore, the paper proposes 4 typical motion models to reduce the complexity of combination. For segments 4-7, they can all be mapped to these 4 types. For 3 segments and below, only the acceleration state or the deceleration state exist, which is easy to determine. Applying the four proposed motion models reduces the complexity of classifications effectively. The content of the section is divided into two parts. The first part is the establishment of the motion model. The second part is the feedrate scheduling based on the motion model.

Establish motion models

For a certain length of L path, 4 special models are analyzed.

Model a: 7-segment S-type acceleration and deceleration

In the case, the given path L is long enough, and the speed curve is a complete 7-segment acceleration and deceleration curve. The maximum feedrate can reach V_m, and the maximum acceleration can reach A_m. The time of each segment satisfies the relational expression $T_1=T_3=T_5=T_7$, $T_2=T_6$. And $T_1, T_2, T_3, T_4, T_5, T_6, T_7$ can be obtained by using Table 1 and formula(4)(5)(6)(7). Formula $J_m*\tau_1*\tau_2+J_m*\tau_1*\tau_1=V_m$ is introduced in Appendix B. The gray represents the value 0 or does not exist in the table. Fig3(a) is the 7-segment S-type acceleration and deceleration.

Model b: 4-segment S-type acceleration and deceleration

In the case, the given path L is not long enough, and the speed curve is a 4-segment acceleration and deceleration curve. This speed curve only includes AA segment (t_0-t_1), DA segment(t_2-t_3), DD segment(t_4-t_5), AD segment(t_6-t_7). The maximum feedrate can reach V_m' and not reach V_m, and the maximum acceleration can reach a_m/A_m. The time of each segment satisfies the relational expression $T_1=T_3=T_5=T_7$, $T_2=T_6=0$, $T_4=0$. And T_1, T_3, T_5, T_7 can be obtained using Table 2 and formula(4)(5)(6)(7). Fig3(b) is the 4-segment S-type acceleration and deceleration.

Model c: 6-segment S-type acceleration and deceleration

In the case, the given path L is not long enough, and the speed curve is a 6-segment acceleration and deceleration curve. This speed curve only includes AA segment (t_0-t_1), UA segment (t_1-t_2), DA segment (t_2-t_3), DD segment (t_4-t_5), UD segment (t_5-t_6), AD segment (t_6-t_7). The maximum feedrate reach V_m/V_m', and the maximum acceleration can reach A_m. The time of each segment satisfies the relational expression $T_1=T_3=T_5=T_7$, $T_2=T_6$, $T_4=0$. And $T_1, T_2, T_3, T_5, T_6, T_7$ can be obtained by using Table 3 and formula(4)(5)(6)(7). Fig3(c) is the 6-segment S-type acceleration and deceleration.

Model d: 5-segment S-type acceleration and deceleration

In the case, the given path L is not long enough, and the speed curve is a 5-segment acceleration and deceleration curve. This speed curve only includes AA segment (t_0-t_1), UA segment (t_1-t_2), DA segment (t_2-t_3), DD segment (t_4-t_5), AD segment (t_6-t_7) or DD segment (t_4-t_5), UD segment (t_5-t_6), AD segment (t_6-t_7), AA segment (t_0-t_1), DA segment (t_2-t_3). Taking the UA segment as an example, the maximum feedrate can reach V_m' which is smaller than V_m, and the maximum acceleration can reach A_m. The time of each segment satisfies the relational expression $T_1=T_3$, $T_5=T_7$, $T_6=0$, $T_4=0$. And T_1, T_2, T_3, T_5, T_7 can be obtained using Table 4 and formula(4)(5)(6)(7). Fig3(d) is the 5-segment S-type acceleration and deceleration.

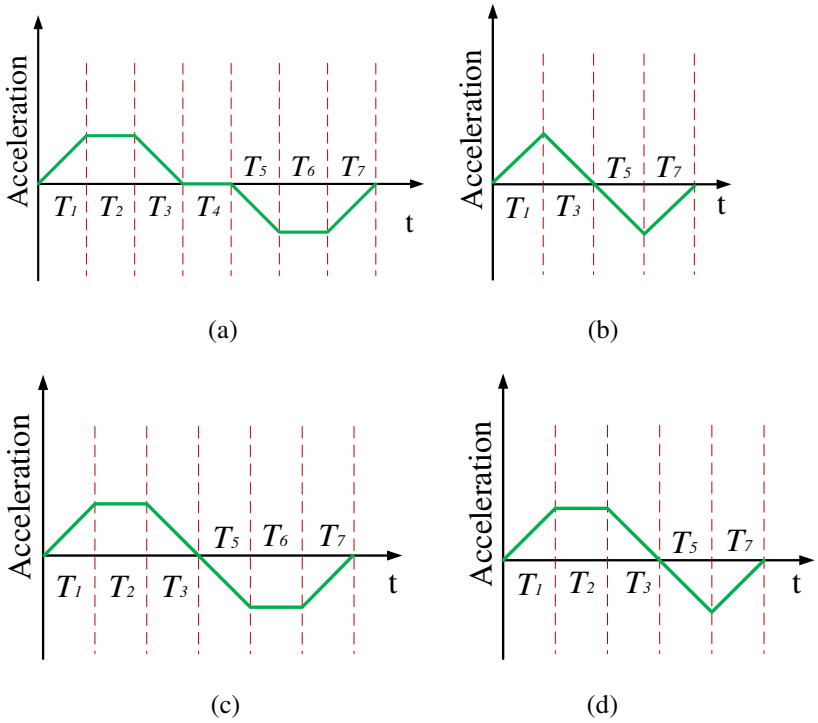

Fig. 2 four motion model. (a) 7-segment S-type acceleration and deceleration motion model. (b) 4-segment S-type acceleration and deceleration motion model. (c) 6-segment S-type acceleration and deceleration motion model. (d) 5-segment S-type acceleration and deceleration motion model.

Table 1 The duration of each segment of 7-segment S-type acceleration and deceleration

7-seg	τ_1	τ_2	τ_3	τ_4	τ_5	τ_6	τ_7
A_m	A_m/J_m		τ_1		τ_1		τ_1
L				$s_7=f(\tau_1, \tau_2, \tau_3)=L$			
V_m		$J_m*\tau_1*\tau_2+J_m*\tau_1*\tau_1=V_m$				τ_2	

Table 2 The duration of each segment of 4-segment S-type acceleration and deceleration

4-seg	τ_1	τ_2	τ_3	τ_4	τ_5	τ_6	τ_7
$a_m(A_m)$	a_m/J_m		τ_1		τ_1		τ_1

Table 3 The duration of each segment of 6-segment S-type acceleration and deceleration

6-seg	τ_1	τ_2	τ_3	τ_4	τ_5	τ_6	τ_7
A_m	A_m/J_m		τ_1		τ_1		τ_1
L		$s_3=f(\tau_1, \tau_2)=L/2$				τ_2	

Table 4 The duration of each segment of 5-segment S-type acceleration and deceleration

5-seg	τ_1	τ_2	τ_3	τ_4	τ_5	τ_6	τ_7
A_m	A_m/J_m		τ_1				
V_m'		$J_m*\tau_1*\tau_2+J_m*\tau_1*\tau_1=V_m'$					
L					$s_7=f(\tau_1, \tau_2, \tau_3)=L$		τ_5

Feedrate scheduling

As shown in Fig. 3, the feedrate scheduling is introduced in the section based on the motion model in detail.

L is the given path length. Model c is a 6-segment acceleration and deceleration curve, and *L6* is defined as the critical displacement when the velocity just reaches the maximum speed V_m. Model d is a 5-segment acceleration and deceleration

curve, and $L5$ is defined as the critical displacement when the acceleration can reach the maximum acceleration A_m but the velocity cannot reach V_m. Model b is a 4-segment acceleration and deceleration curve, and $L4$ is the critical displacement when the acceleration just reaches the maximum acceleration A_m.

Category1: When $L6<L<L7$ or $L>L7$, it is a complete acceleration and deceleration curve of 7 segments. $\tau1$-$\tau7$ can be solve using Table 1.

Category2: When $L5<L<L6$, $L6$ is the displacement when the velocity just reaches the maximum speed V_m, so the displacement L only reaches the speed V_m' which is smaller than V_m. Because the displacement $L5$ must be able to reach the maximum acceleration A_m, the displacement L must be able to reach the maximum acceleration A_m. Consequently, L belonging to the interval [$L5$, $L6$] is enough long to reach A_m but not enough long to reach V_m. Therefore, category2 belongs to model c and (τ_1, τ_2, τ_3, τ_4, τ_5, τ_6, τ_7) is solved by the formula in Table 3. For example, τ_1 can be solved by formula A_m/J_m and τ_2 can be solved by formula$s_3=f(\tau_1, \tau_2)=L/2$.

Category3: When $L4<L<L5$, because the displacement $L4$ is just length to reach A_m, the displacement L must be able to reach the maximum acceleration A_m. The displacement $L5$ is enough long to reach A_m but not enough long to reach the maximum speed V_m. Consequently, L belonging to the [$L4$, $L5$] interval is enough long to reach A_m but not enough long to reach V_m. Thereby, category3 belongs to model d and (τ_1, τ_2, τ_3, τ_4, τ_5, τ_6, τ_7) can be solved by the formula in Table 4. τ_1 is solved by formula A_m/J_m, τ_2 is solved by formula $J_m*\tau_1*\tau_2+J_m*\tau_1*\tau_1=V_m'$, and$\tau_3$ is solved by formula $s_7=f(\tau_1, \tau_2, \tau_3)=L$.

Category4: When $L<L4$, because $L4$ is the displacement just reaching the maximum acceleration A_m. L belonging to the [0, $L4$] interval is only enough to reach a_m which is smaller than A_m. Therefore, category4 belongs to model b. (τ_1, τ_2, τ_3, τ_4, τ_5, τ_6, τ_7) can be solved by using formula in Table 2 and τ_1 can be solved by using A_m/J_m.

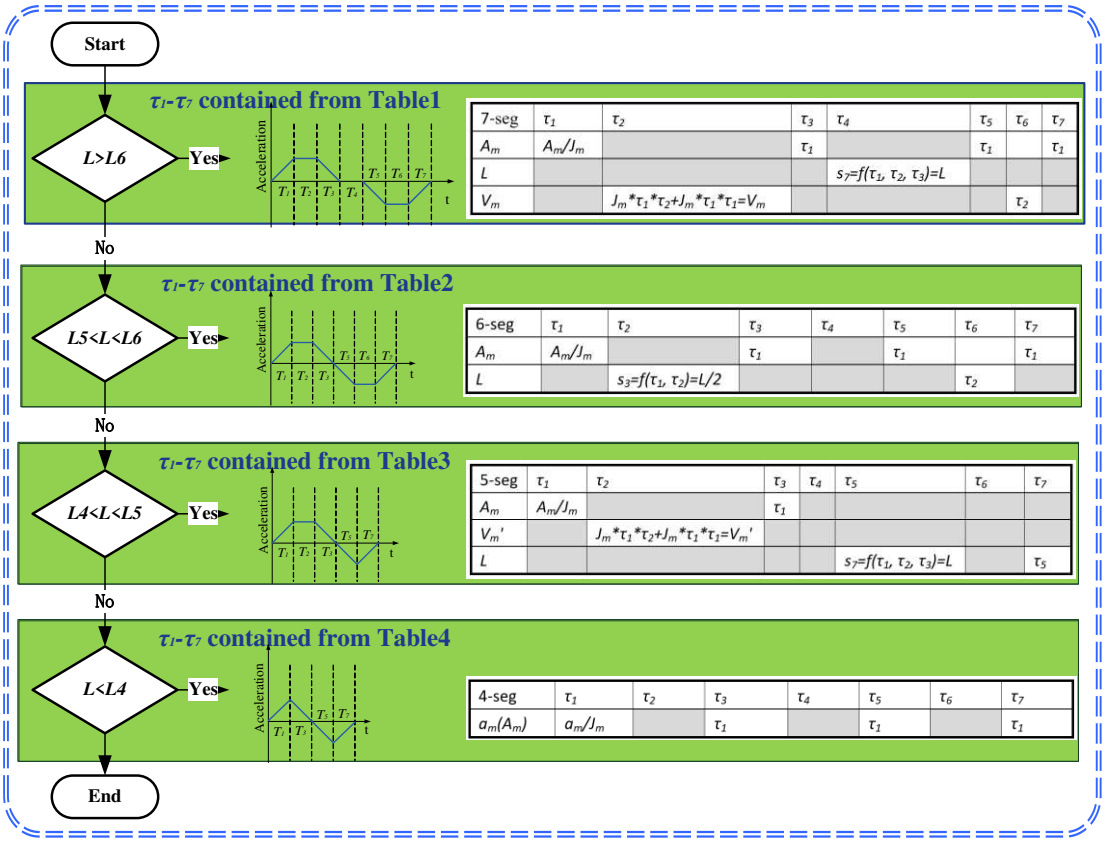

Fig. 3 The relationship between τ_1-τ_7 and path length L in feedrate scheduling. L is the length of given path, and $L6$ is defined as the critical displacement just reaching the maximum speed V_m in Model c. $L5$ is defined as the critical displacement that can reach the maximum acceleration A_m but cannot reach V_m in Model d, and $L4$ is defined as the critical displacement just reaching the maximum acceleration A_m in Model b.

2.3 Backward and forward planning

Backward planning

Path planning is divided into backward planning and forward planning [27]. Backward planning is introduced in the part,

and forward planning is introduced in next part. The purpose of backward planning is to obtain the maximum starting speed within a limit of sensitive speed area. The paths are planned in groups not one by one, and how to get the maximum starting speed with the end speed under the feedrate constraint [30, 32]. This is the meaning of backward planning. The backward planning process is as follows:

Step1: getting the end speed V_e of last path.

Step2: For the last path with a given length of L_n, V_e is regard as the starting speed V_{s1}. The last path is planned reversely to obtain the end speed V_{e1} with the feedrate scheduling introduced in previous section.

Step3: V_{es2} is the smallest velocity between the end velocity V_{e1} and the sensitive velocity V_{sensor}. For the last path with a given length of L_{n-1}, V_{es2} is the starting speed of L_{n-1}. L_{n-1} is planned reversely to obtain the end speed V_{e2}.

Step4: step4 repeat step2 until the first path with a given length of L_1 is planned reversely and the end speed V_{en} is obtained. This end speed V_{en} is the maximum starting speed V_s of the set of paths.

Forward planning

In the previous section, the starting velocity V_s of a set of paths is obtained by backward planning. With V_s planned by the backward planning and the feedrate scheduling in Table 1-Table 4, the durationτ_1, τ_2, τ_3, τ_4, τ_5, τ_6, τ_7 of each segment of the 7-segment S-type acceleration and deceleration can be obtained for a given length L [33][34].

3. Interpolation pre-compensation based on discrete inverse transfer function

Dispersed micro-segment from CAM is input to the CNC system. The CNC system receives micro-segment from CAM and transmits the commanded position P_r to the drive system. The output of the drive system is the actual position P_a. As shown in Fig. 4.

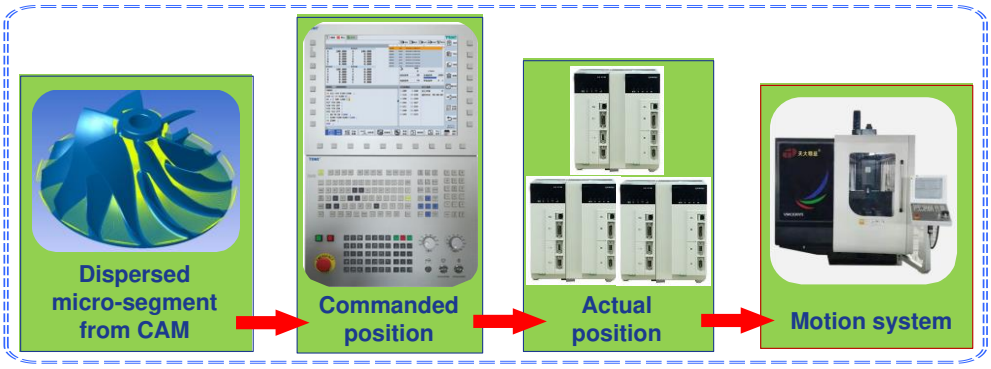

Fig. 4 Flow diagram of interpolation position

The commanded position P_r output by the CNC system can accurately express the contour contour of the workpiece, but the actual position P_a which is output from the drive system cannot completely follow the commanded position P_r. As a result, the actual position P_a and the commanded position P_r are not exactly same, leading to a reduction in machining accuracy [35]. This machining deviation is called contour error[36]-[40], as shown in Fig. 5.

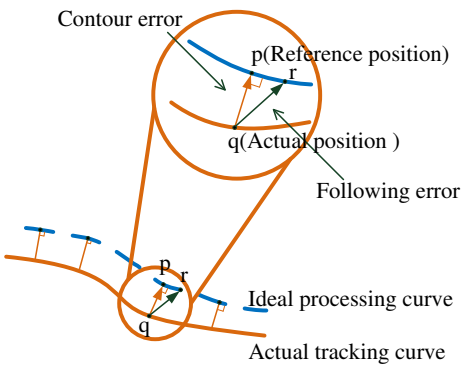

Fig. 5 Contour error, where q is the actual position and r is the reference position and p is the point closest to q in the reference curve. The distance between q and p is the contour err. The distance between q and r is the following error.

This difference is caused by the unsteady response of the drive system, which is a transient state before steady-state response of drive system. The paper proposes interpolation pre-compensation theory to avoid contour errors caused by unsteady response. In Section 4.1, the cause of contour error is analyzed. In Section 4.2, Before proposing the pre-compensation theory, transfer function discretization theory is induced at first, and pre-compensation theory is introduced in section 4.3.

3.1 Response analysis of drive system

The drive system is consisted by the closed-loop transfer function *G(s)* [44]. For a typical drive system is composed by a position loop and a speed loop and PMSM mathematical model. the control block diagram is shown in Fig. 6 [45]. The position loop is controlled by the P controller, and the velocity loop is controlled by the PI controller. In this situation, the PMSM can be reasonably approximated by a first-order inertial system (FOIS).

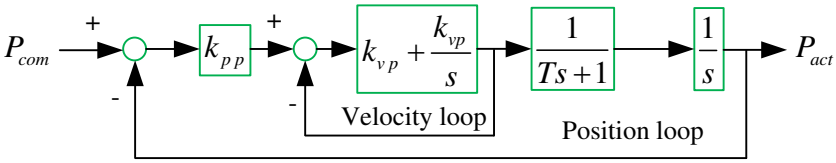

Fig. 6 Block diagram of the typical drive system

Therefore, the closed-loop transfer function *G(s)* of drive system can be calculated as

$$G(s)=\frac{k_{pp}k_{vp}k_{vi}}{Ts^3+\left(1+Tk_{vp}k_{vi}\right)s^2+k_{vp}k_{vi}s+k_{pp}k_{vp}k_{vi}} \tag{12}$$

The step response of the closed-loop transfer function *G(s)* is depicted in Fig. 7. From Fig. 7, it is easy to find that, before reaching a steady-state, the FOIS with time constant T include a low bandwidth zone about (3~4) T. The low bandwidth zone leading to contour error is presented obviously in high-precision processing field. T is generally greater than interpolation period T_s. Therefore, it is difficult, even impossible, to reach steady-state within T_s. To address this issue, PMSM is simplified a first-order inertial system in Ref [45, 46], while the simplify model is not the real mathematical model. Model error is introduced by the simplify model inevitably. A new model is proposed by the paper to the eliminate the model error.

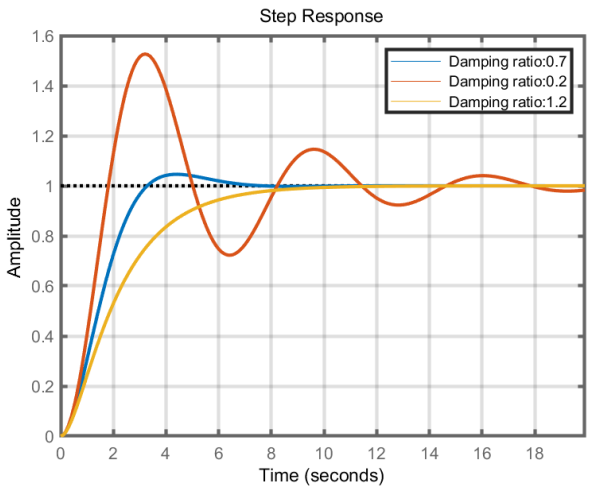

Fig. 7 Unsteady-to-steady-state response of drive system

3.2 Transfer function discretization theory

The response of transfer function from unsteady-state to steady-state causes contour errors. Therefore, the article focuses on the unsteady-state process of transfer function to solve the issue. Before discussing the issue, a brief explanation of transfer function discretization theory is induced at first.

The CNC system is a linear time invariant systems (LTI) [49]. The response of LTI system is:

$$y(t)=\int x(t)\cdot h(t)dt \tag{13}$$

h(t) is the transfer function. *x(t)* is the input signal of transfer function. *y(t)* is the output signal of transfer function. The key parameter of CNC system is interpolation period T_s. The CNC system transmits the commanded position P_r with T_s, periodically. Hence, the essence of the CNC system is a discrete time system, not a continuous time system. The smaller the interpolation period, the higher the interpolation accuracy. No matter how small the interpolation period T_s is, the series of commanded position still are dispersed. Based on the analysis, *x(t)* should be replaced by *x(n)*. The form of formula(13) should be:

$$y(t)=x(n)*h(t) \tag{14}$$

However, the drive system which is a transfer function is a continuous time system *h(t)*. It can be seen from the above formula(14), the transfer function *h(t)* can be degenerated into a discrete-time system *h(n)*. The form of formula(14) should be:

$$y(n)=x(n)*h(n) \tag{15}$$

Theory 1: The transfer function is the ratio of the output function to the input function in Z-transform[47], which is $H(\mathrm{z})=\dfrac{Y(\mathrm{z})}{X(\mathrm{z})}$. If the input function is a unit pulse, the Z-transform of unit pulse is 1. Thereby, the transfer function is the output in Z-transform, which is $H(\mathrm{z})=Y(\mathrm{z})$.

Using the Theory 1 and formula(15), the transfer discrete function *h(n)* can be obtained.

Theory 2: Delay, multiplication, and addition are the basic operations of LTI.

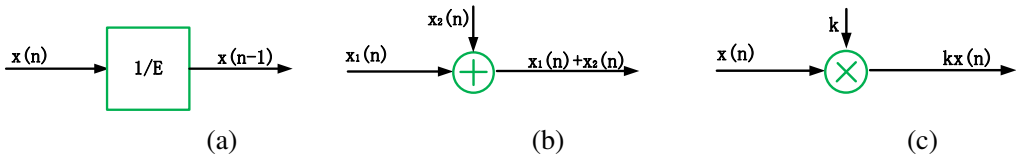

Fig. 8 The basic operations of LTI system. (a) The delay of LTI system. (b) The addition of LTI system. (c) The multiplication of LTI system.

Using Theory2, the serious of commanded position $P_r(n)$ can be regarded as a serious of discrete sequence with different amplitudes and different time-shift. The discrete transfer function theory is supported by Theory 1 and Theory 2.

3.3 Discrete inverse transfer function algorithm

In section 3.2, the conclusion that the transfer function which is the radio of output and input in Z-transform is the response of the unit pulse is introduced. In Section 3.1, the transition process from the unsteady-state response to the steady-state response leads to contour errors. To address the issue, the theory of discrete inverse transfer function is proposed.

The drive system response can be divided into two parts, the steady-state response and the unsteady-state response. From the conclusion of Section 3.2, the discrete transfer function ***H(m)*** is the unsteady-state response of unit pulse and the unit pulse response of steady-state is 0. The length of unsteady response is *m*. Therefore, ***H(m)*** is the discrete transfer function of drive system. Besides, *h(0)* is the 0 time response of the transfer function, the response of the transfer function has not yet started, so *h(0)*=0.

$$\boldsymbol{H(m)}=\left(h(1),h(2),h(3)\mathrm{L}\ \ h(m)\right)^T \tag{16}$$

For the law of conservation of energy, the sum of the output should be a unit pulse if the input is a unit pulse. However, the sum of the output is not a unit pulse because of the data accuracy error, the ***H(m)*** should be normalized.

$$\boldsymbol{H(m)} = \left(\frac{h(1)}{\sum_{i=0}^{m} h(i)}, \frac{h(2)}{\sum_{i=0}^{m} h(i)}, \frac{h(3)}{\sum_{i=0}^{m} h(i)} \mathrm{L} \frac{h(m)}{\sum_{i=0}^{m} h(i)} \right)^T \tag{17}$$

P_r ,P_a and P_c represent the commanded position, the actual position and the pre-compensated position. Generally, the commanded position P_r is transmit to the drive system which is a transfer function, and the actual position P_a is output. Based on the analysis of the response of transfer function, P_a is not in accordance with P_r. If the consistency is achieved between P_a and P_r after processed by ***H(m)***, P_r must be compensated and P_c is the pre-compensated position. That is, the commanded position P_r is replaced by the pre-compensated position P_c, and actual position P_a is replaced by the commanded position P_r. The pre-compensation principle for the commanded position will be introduced in detail in the section.

If a constant value is continuously input to the transfer function, it is always a steady-state response after a while. Only when the input changes, will the unsteady response appear. Therefore, the concept of input increment ***W(n)*** which is named weight is introduced.

$$\boldsymbol{W(n)} = \begin{pmatrix} w(1) \\ w(2) \\ w(3) \\ \mathrm{M} \\ w(m) \\ w(m+1) \\ \mathrm{M} \\ w(n) \end{pmatrix} = \begin{pmatrix} P_r(1) - init \\ P_r(2) - init \\ P_r(3) - init \\ \mathrm{M} \\ P_r(m) - init \\ P_r(m+1) - init \\ \mathrm{M} \\ P_r(n) - init \end{pmatrix} = \begin{pmatrix} P_r(1) - P_r(0) \\ P_r(2) - P_r(0) \\ P_r(3) - P_r(0) \\ \mathrm{M} \\ P_r(m) - P_r(0) \\ P_r(m+1) - P_r(0) \\ \mathrm{M} \\ P_r(n) - P_r(0) \end{pmatrix} \tag{18}$$

The input increment ***W(n)*** is the increment of the current commanded position $P_r(i)$ relative to the first commanded position $P_r(0)$ which is the initial value. Based on the above analysis of pre-compensation principle, ***W(n)*** is the output of the transfer function. Similarly, ***PW(n)*** is the increment of pre-compensation position and input signal.

$$\boldsymbol{PW(n)} = \begin{pmatrix} pw(1) \\ pw(2) \\ pw(3) \\ \mathrm{M} \\ pw(m) \\ pw(m+1) \\ \mathrm{M} \\ pw(n) \end{pmatrix} = \begin{pmatrix} P_c(1) - init' \\ P_c(2) - init' \\ P_c(3) - init' \\ \mathrm{M} \\ P_c(m) - init' \\ P_c(m+1) - init' \\ \mathrm{M} \\ P_c(n) - init' \end{pmatrix} = \begin{pmatrix} P_c(1) - P_c(0) \\ P_c(2) - P_c(0) \\ P_c(3) - P_c(0) \\ \mathrm{M} \\ P_c(m) - P_c(0) \\ P_c(m+1) - P_c(0) \\ \mathrm{M} \\ P_c(n) - P_c(0) \end{pmatrix} \tag{19}$$

Then under the discrete inverse transfer function, the relationship between input ***PW(n)*** and output ***W(n)*** is

$$\begin{pmatrix} pw(1) & 0 & 0 & \mathrm{L} & 0 \\ pw(2) & pw(1) & 0 & \mathrm{L} & 0 \\ pw(3) & pw(2) & pw(1) & \mathrm{L} & 0 \\ \mathrm{M} & \mathrm{M} & \mathrm{M} & \mathrm{O} & \mathrm{M} \\ pw(m) & pw(m-1) & pw(m-2) & \mathrm{L} & pw(1) \\ pw(m+1) & pw(m) & pw(m-1) & \mathrm{L} & pw(2) \\ \mathrm{M} & \mathrm{M} & \mathrm{M} & \mathrm{O} & \mathrm{M} \\ pw(n) & pw(n-1) & pw(n-2) & \mathrm{L} & pw(n-m+1) \end{pmatrix} \cdot \begin{pmatrix} h(1) \\ h(2) \\ h(3) \\ h(4) \\ h(5) \\ h(6) \\ \mathrm{M} \\ h(m) \end{pmatrix} = \begin{pmatrix} w(1) \\ w(2) \\ w(3) \\ \mathrm{M} \\ w(m) \\ w(m+1) \\ \mathrm{M} \\ w(n) \end{pmatrix} \tag{20}$$

Then the pre-compensation position $P_c(i)$ is inputted to the drive system is:

$$\boldsymbol{PW(n)} = \begin{pmatrix} w(1)/h(1) \\ \dfrac{w(2)}{h(1)} \\ \dfrac{w(3)}{h(1)} \\ \vdots \\ \dfrac{w(m)}{h(1)} \\ \dfrac{w(m+1)}{h(1)} \\ \vdots \\ \dfrac{w(n)}{h(1)} \end{pmatrix} - \begin{pmatrix} 0 \\ \dfrac{pw(1)\cdot h(2)}{h(1)} \\ \dfrac{pw(2)\cdot h(2)+pw(1)\cdot h(3)}{h(1)} \\ \vdots \\ \dfrac{pw(m-1)\cdot h(2)+\cdots+pw(1)\cdot h(m)}{h(1)} \\ \dfrac{pw(m)\cdot h(2)+\cdots+pw(2)\cdot h(m)}{h(1)} \\ \vdots \\ \dfrac{pw(n-1)\cdot h(2)+\cdots+pw(n-m+1)\cdot h(m)}{h(1)} \end{pmatrix} \tag{21}$$

4. Simulation experiments

In the section, simulation experiments are used to evaluate the performance of the proposed PDIT. FOIS used as a comparison is a method to improve the contour accuracy, the same as CEEC. FOIS is approximated mathematic pattern of drive system simplified by a first-order inertial system. The pattern composed of a position loop and a speed loop and PMSM mathematical model. The theory contour error is deduced by the simplified FOIS and compensated to the commanded position. In CEEC, Newton's method is used to analyze the contour error, and the iterative learning control(ILC) [50] method is used to compensate the contour error. In order to avoid mechanical precision interference such as bearing clearance, simulation experiment is adopted to evaluate the performance of the proposed PDIT, and a simplified drive system model based on the simulink module is employed [51, 54]. Fig. 9 is a simplified drive system.

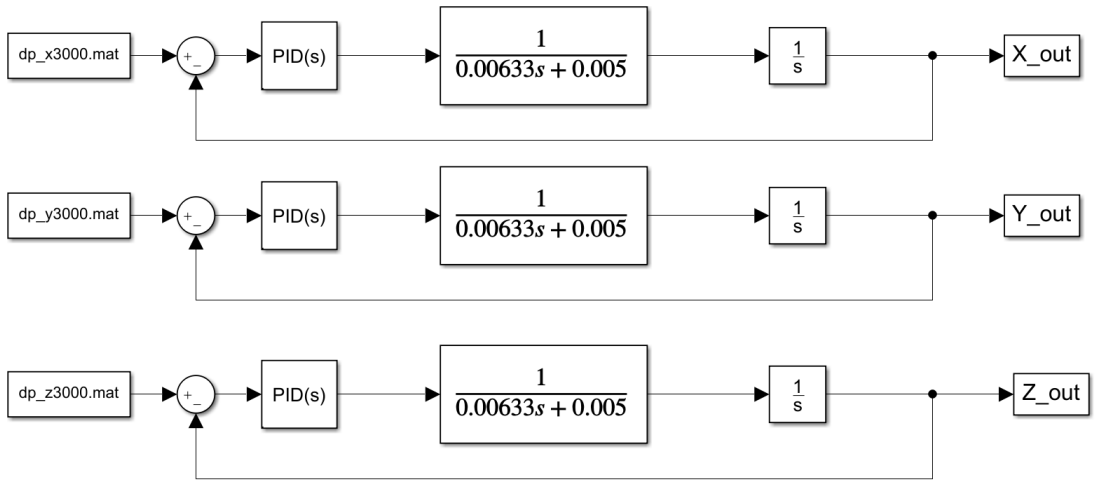

Fig. 9 The mathematical simplified model of drive system

Since the same method is applied to the XYZ axis, take the X axis as an example to expand the introduction. In the X axis, the K_p, K_i, *and* K_d parameters are 1200, 300, and 5 respectively.

A butterfly and spiral graphics are used for simulation experiments to verify the PDIT proposed in this paper. The butterfly was used in the first simulation experiment, and the spiral was used in the second simulation experiment. The parameter interpolation based on Taylor expansion and the feedrate scheduling in Chapter2 are used to obtain the position $C(u_{i+1})$ of the next interpolation cycle. The PDIT proposed in Chapter3 is used to pre-compensate the position $C(u_{i+1})$ of the next interpolation cycle.

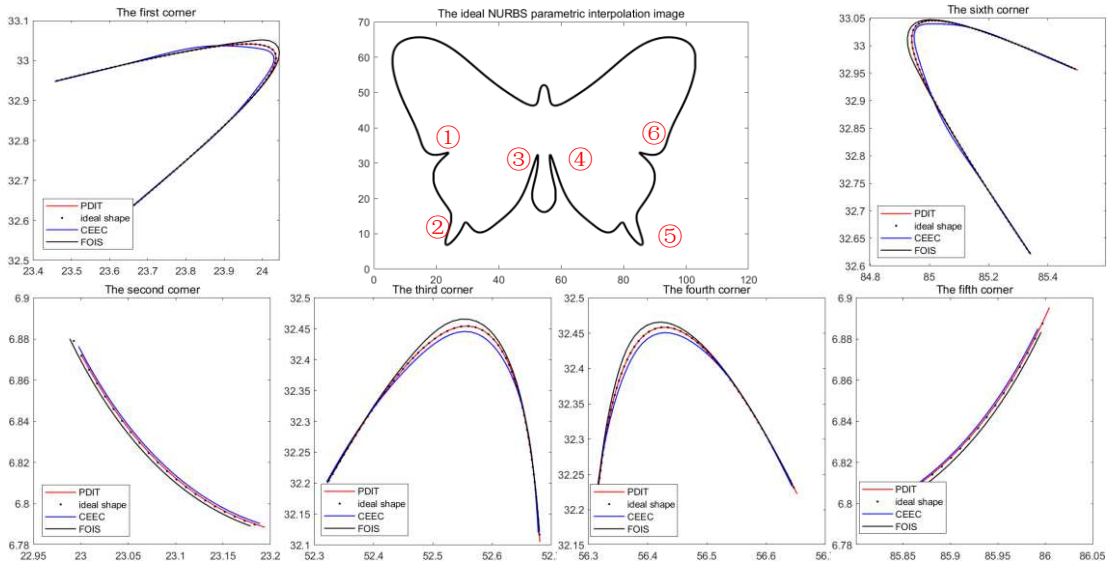

Fig. 10 Evaluation of the contour error of the butterfly's corner

In Fig. 10, the improvement of the butterfly corner in the six corners, marked 1-6 in the figure, is experimentally verified. In the figure, the red trajectory is the PDIT theory proposed by the paper, the black trajectory is the FOIS method, and the blue trajectory is the CEEC theory. The black point is the ideal commanded position, which is used as a reference for comparing the PDIT, FOIS and the CEEC theory. In Fig. 10, at the corner of the butterfly trajectory, it can be clearly seen that the PDIT theory is closer to the ideal interpolation curve than the CEEC theory. The CEEC trajectory is closer than the FOIS trajectory.

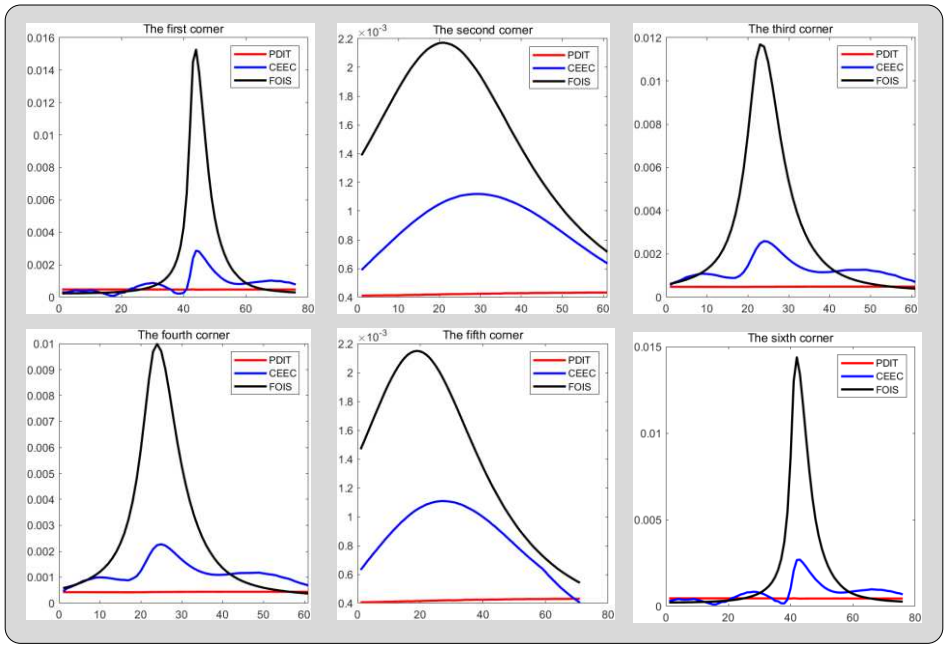

Fig. 11 Datasheet of the contour error of the butterfly's corner

At the same time, in Fig. 11, the contour error is given for all the six corners. The contour error of PDIT theory is almost a horizontal curve, while the CEEC and FOIS theory is a curve that fluctuates up and down. The contour error of PDIT theory is significantly smaller than that of CEEC theory, and the contour error of corner in CEEC is smaller than the FOIS. The contour error analysis of six corner is give in Table 5 with PDIT FIOS and CEEC. The maximum contour error level of PDIT theory is 0.4um, and the maximum contour error level of CEEC theory is 2.9um, the FOIS is 15um. The interpolation pre-compensation theory based on the discrete inverse transfer function proposed in this paper has a significant improvement effect in terms of contour.

Table 5 The contour error analysis of six corner in different methods

Compensation mathod	PDIT			FOIS			CEEC		
Contour error	Maximum	Mean	Minimum	Maximum	Mean	Minimum	Maximum	Mean	Minimum

The 1st corner	0.000484	0.00047	0.000464	0.015263	0.0020544	0.000238	0.002873	0.0008215	0.00008
The 2nd corner	0.000433	0.00042392	0.000412	0.002172	0.00156605	0.000716	0.001119	0.0009185	0.00059
The 3rd corner	0.000491	0.000485	0.000477	0.011689	0.00288815	0.000394	0.002595	0.0012704	0.000561
The 4th corner	0.000442	0.00043371	0.000422	0.009988	0.00259823	0.000373	0.002272	0.0011701	0.000462
The 5th corner	0.000431	0.00042125	0.000407	0.00215	0.0014043	0.000542	0.001109	0.0008504	0.000404
The 6th corner	0.000457	0.00044999	0.000438	0.014388	0.00195746	0.000217	0.002691	0.0007969	0.000104

In the second experiment, take the spiral curve as an example. The spiral curve formula (22) is as follows [16]. In Fig. 12, the red trajectory is the PDIT theory proposed in this article, the blue trajectory is the CEEC theory, and the black point is the ideal commanded position. Fig. 12 shows that the red trajectory of the PDIT theory is closer to the ideal curve than the blue curve of the CEEC theory. Fig. 13 and Table 6 are the contour error of PDIT theory and CEEC theory. The contour error of PDIT is smaller than CEEC. For the contour error of the spiral curve, the contour error of PDIT is 0.000849-0.147475*mm*, and the contour error of CEEC is 0.001189-0.496462*mm*.

$$\begin{cases} c_x(t) = cos(2\pi t)t \\ c_y(t) = 3sin(2\pi t)t \end{cases} \tag{22}$$

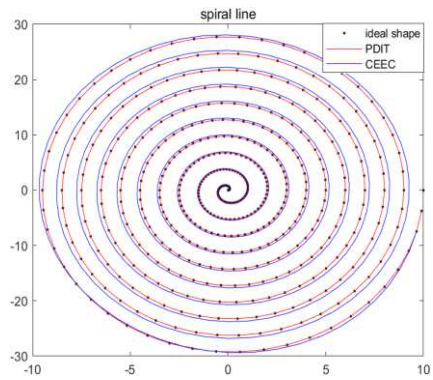

Fig. 12 The evaluation of spiral curve

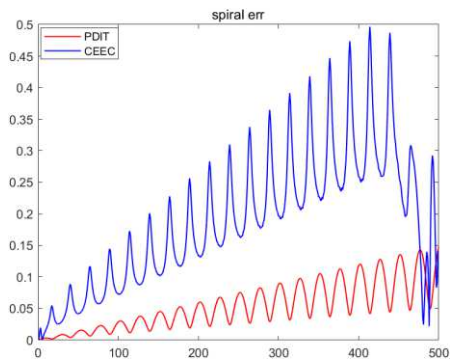

Fig. 13 Contour error of the entire path of the spiral curve

Table 6 Datasheet of the contour error of the spiral curve

Spiral error			
Theory	Maximum	Minimum	Delta
PDIT	0.147475	0.000849	0.146626
CEEC	0.496462	0.001189	0.495273

5. Conclusions

The PDIT is proposed by the paper and able to effectively improve the machining accuracy by reducing contour, especially at the corners. Without decreasing feedrate, the contour error is reduced by PDIT with the consistent of commanded position and the actual position. In Chapter 5, the butterfly and spiral pattern are simulated to evaluate the

performance of PDIT. It can be clearly seen from the Fig. 10 and Fig. 12 that PDIT has a significant effect on improving machining accuracy. The commanded position with PDIT can completely avoid the influence of unsteady response, which is helpful to realize high-speed and high-precision machining in CNC field. The PDIT only contains basic operations of addition, subtraction, multiplication and division without any complex operations such as integration, which does not bring burden to CPU and hardly affects the real-time performance of NURBS interpolator.

Appendix:

Appendix A:

In the acc-acceleration segment, the jerk is a constant positive value, and the acceleration increases at a constant value. In the uniform acceleration segment, the value of the jerk is 0, and the acceleration reach a maximum value A_m. In the dec-acceleration segment, the jerk is a constant negative value, and the acceleration decreases at a constant value. In the constant speed segment, the values of jerk and acceleration are both 0, and the velocity reach the maximum value V_m. In the dec-deceleration segment, the jerk is a constant negative value, and the acceleration decreases at a constant value. In the uniform deceleration segment, the value of jerk is 0, and the acceleration reach a minimum value $-A_m$. During the acc-deceleration segment, the jerk is a constant positive value, and the acceleration increases at a constant value until it reaches zero.

t_i represents the critical moment in segments, so t_1-t_7 are the end moments of the first to seventh segments. T_i represents the duration of each segment, so $T_1=t_1-t_0$ is the duration of the first segment, and $T_7=t_7-t_6$ is the duration of the seventh segment. Others are the same. τ_i represents the time difference between the any moment in one segment and the starting moment of the same segment. For example, in the interval $t_6<t<t_7$, $\tau_7=t-t_6$. The acceleration in each segment is represented by $a_i(t)$, so $a_1(t)$-$a_7(t)$ is the acceleration of the first to seventh segments. The acceleration at the end of each segment is represented by a_i, so a_1-a_7 are the final accelerations of the first to seventh segments. The speed in each segment is represented by $v_i(t)$, so $v_1(t)$-$v_7(t)$ is the speed of the first to seventh segments. The speed at the end of each segment is represented by v_i, so v_1-v_7 are the final speeds of the first to seventh segments. The displacement in each segment is represented by $s_i(t)$, so $s_1(t)$-$s_7(t)$ is the displacement in the first to seventh segments. The displacement at the end of each segment is represented by s_i. The total displacement of the entire 7 segments is represented by S_{total}.

Appendix B:

In model a, After the first three segments, which are AA, UA and DA, the speed increases from 0 to the maximum speed V_m. The speed increment in the AA segment is Δv_1. The speed increment in the UA and AD segment is Δv_2 and Δv_3, individually. And there is a relation: $t_1=T_1$, $t_2=T_1+T_2$, $t_3=T_1+T_2+T_3$, $T_1=T_3=T$.

0-t_1:

$$\mathrm{V}v_1=\int_0^{T_1} \mathrm{a}dt=\frac{1}{2}J_m t^2\,\Big|_0^{T_1}=\frac{1}{2}J_m T_1^2 \quad \text{(C.1)}$$

t_1-t_2:

$$\mathrm{V}v_2=\int_{t_1}^{t_2} \mathrm{a}dt=\int_{t_1}^{t_2} J_m T_1 dt=J_m T_1 t\,\Big|_{t_1}^{t_2}=J_m T_1 T_2 \quad \text{(C.2)}$$

t_2-t_3:

$$a=J_m T_1-J_m(t-t_2) \quad \text{(C.3)}$$

$$\mathrm{V}v_3=\int_{t_2}^{t_3} adt=\int_{t_2}^{t_3} J_m T_1-J_m(t-t_2)dt=\frac{1}{2}J_m T_1^2 \quad \text{(C.4)}$$

0-t_3:

$$\mathrm{V}v=\mathrm{V}v_1+\mathrm{V}v_2+\mathrm{V}v_3=\frac{1}{2}JT_1^2+J_m T_1 T_2+\frac{1}{2}JT^2=JT_1^2+J_m T_1 T_2 \quad \text{(C.5)}$$

Funding
This work was supported by the National Natural Science Foundation of China [grant number 51975402] and Tianjin Science and Technology Committee [project number 20YDTPJC01730].

Conflicts of interest/Competing interests
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Availability of data and material
Not applicable

Code availability
Not applicable

Authors' contributions
Libo Cao: Conceptualization, Software, Validation, Formal analysis, Data curation, Writing - original draft, Writing - review & editing. Taiyong Wang: Conceptualization, Funding acquisition, Supervision, Resources. Yongbin Zhang: Methodology, Investigation, review , Data curation. Jingchuan Dong: Conceptualization. Songhui Jia: Methodology, Investigation, Writing - review & editing. Chong Tian: Data curation, Investigation. Qingjian Liu: Funding acquisition.

Ethics approval
Approval

Consent to participate
Consent

Consent for publication
Consent

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